
Physics-Posterior Distillation for Trustworthy Methane Source Attribution under Time-Varying Winds

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Abstract

Rapid repair of oil-and-gas methane super-emitters is among the highest-leverage near-term climate actions, but it is bottlenecked not by detection (satellites already flag plumes) but by trustworthy *attribution*: turning a detection into a source location a crew will act on. A neural localizer over a sparse mast network can output a region, and split conformal prediction can make that region provably contain the source 90% of the time — yet a calibrated region can still be far larger than the physics permits. We study this amortization gap on CH₄-T, a methane-configured testbed (500 m site, 3-D Gaussian plume with Pasquill–Gifford stability, kg/h releases, ppm masts) driven by a realistic *time-varying measured wind*, so that time-of-arrival information is available exactly as in real continuous monitoring. Because the forward model is closed-form and linear in the release rate, we compute the *exact* Bayesian posterior of every scenario and use it twice: as a physics-derived training target, and as the reference that measures how close a learned localizer comes to the information limit. A standard DeepSets localizer is well-calibrated (0.89) but its median 90% region has a 94 m equivalent radius against a 7 m Bayes limit. Distilling the exact posterior — same architecture, data, and optimizer, only the training target changes — tightens it to 74 m (paired $p < 10^{-60}$; stable across three seeds; raw untempered targets fail). Read against the U.S. EPA 50 m super-emitter actionability radius, this is meaningful movement toward decision-grade attribution at no inference cost.

1 Introduction

Methane warms the climate roughly $80\times$ more than CO₂ per ton over 20 years and clears the atmosphere in about a decade, so cutting it slows warming within our lifetimes; a disproportionate share comes from a small number of fixable oil-and-gas *super-emitters* [U.S. EPA, 2024; European Union, 2024]. Detection is increasingly solved — UNEP’s satellite alert system issued 1,200 notifications in two years — but only 1% were acted on [UNEP, 2024]. A key reason is *attribution*: a satellite gives coordinates over an industrial area with many candidate sources, and someone must say which facility, with enough confidence to dispatch a crew or, under the EPA Super-Emitter Program, to attach reporting obligations to an operator within 50 m of the reported location.

A learned localizer over a sparse network of fixed methane masts can turn concentration readings into a heatmap of source locations, and split conformal prediction can convert that heatmap into a region that provably contains the true source with probability 90%. But coverage is a statement about *frequency*, not *efficiency*: a region can be honestly 90% and still cover half the site, sending crews to search far more area than the data warrant. What is missing is a way to ask whether a calibrated region is as *small* as the physics allows — and, if not, to make it smaller.

Our observation is that when the training data come from a tractable simulator, as they do throughout simulation-based inference, the complete Bayesian posterior of every scenario can be computed offline and used two ways: as a dense, physics-derived *teacher* that supervises what the uncertainty should look like, and as the exact *reference* that measures how much of the achievable sharpness a learned localizer recovers. We instantiate this on a methane-configured testbed with realistic time-varying winds and show that changing only the training target moves a standard localizer’s median search region from a 94 m to a 74 m radius against a 7 m Bayes limit.

Contributions. (1) A methane-configured attribution testbed, CH4-T, with time-varying measured winds and exact per-scenario posteriors. (2) An exact-posterior audit that reports calibrated region size in physical units against the information limit and the regulatory radius. (3) Physics-posterior distillation: supervising an unchanged localizer with the exact posterior substantially sharpens its regions at no inference cost, provided the teacher is mildly tempered (raw targets fail). We claim no new architecture, no new conformal algorithm, no coverage under distribution shift, and no field deployability; CH4-T is a controlled testbed (steady release, flat terrain, known stability, measured wind), and the path to deployment runs through controlled-release recalibration.

2 CH4-T: a methane-configured testbed with time-varying winds

Physics. A source of rate Q (kg/h) at $\mathbf{x}_s$ on a 500×500 m site emits a steady 3-D Gaussian plume with ground reflection and Briggs open-country dispersion under a Pasquill–Gifford stability class (A–F, known). The novel ingredient is that each scenario has a *known, fluctuating wind sequence* $\mathbf{u}(t_k)$, $k = 1, \dots, 30$ — a mean wind (1–8 m/s) plus an AR(1) meander in direction (± 20 – 40°) and speed — exactly the wind that on-site anemometry measures in practice. The plume responds quasi-statically, so the ppm reading at mast i and time t_k is

$$y_{ik} = Q g(\mathbf{x}_i; \mathbf{x}_s, \mathbf{u}(t_k), \text{class}) + \varepsilon_{ik}, \quad \varepsilon_{ik} \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

with g the closed-form unit-rate plume (0.2–3 ppm noise, Bernoulli dropout, 4–12 masts). As the wind veers over the window, the plume sweeps a fan across the masts (Figure 2, left), so *when* each mast lights up — correlated with the measured wind — localizes the source, exactly as time-of-arrival does in real continuous monitoring. This is decisive: with a *constant* wind the same plume misses randomly placed masts and 57% of scenarios are un-localizable; with the time-varying wind only 6% are.

Exact posteriors. Because g is closed-form and y is linear in Q , the per-cell response over the retained (mast, time) pairs gives a Gaussian likelihood; we marginalize Q over a LogUniform(10, 500) prior in closed form and normalize over a 64×64 grid to obtain the exact posterior $p(c | \mathbf{y})$. Sources lie on the grid (cell-level attribution): with time-of-arrival the posterior is often sharper than a cell, and a sub-cell source offset would otherwise leave the true cell nearly empty and break the audit. The exact posterior is used as both teacher and reference. A blocking gate requires conformal-on-exact-posterior to self-calibrate (0.95 here); an argument-order bug that broke it was caught by this gate.

3 Method: three models, one architecture

The localizer is a permutation-invariant DeepSets network (4.87M parameters) over reading tokens [Fourier(x_i, y_i, t_k), asinh(y_{ik}/σ), $\mathbf{u}(t_k)$] with masked mean+max pooling and a stability/noise/mast-count context, decoding a softmax over the 64^2 grid. Training minimizes cross-entropy to a target; **M0** uses the standard smoothed one-hot at the true cell, **M1** uses the scenario’s exact posterior, tempered by a 0.75-cell Gaussian, with the target grid permuted under the same D4 symmetry as the (physics-equivariant) input augmentation. Everything else — architecture, 10k training scenarios, AdamW, schedule — is identical between M0 and M1; only the supervision target changes. Split conformal (randomized highest-density-mass, computed in numerically safe tail-mass form) wraps every model at verified 90% coverage, so region-size differences isolate sharpness from calibration.

Table 1: CH4-T test results ($n = 2000$; median 90% region as equivalent radius in meters; three-seed range). Both models are conformally calibrated; the target is the only difference between M0 and M1.

	M0 (standard)	M1 (distilled)	Raw-teacher	Bayes limit
Coverage @ 90%	0.886–0.905	0.896–0.905	0.899	0.95
Region radius (m)	94–98	73–74	134	7

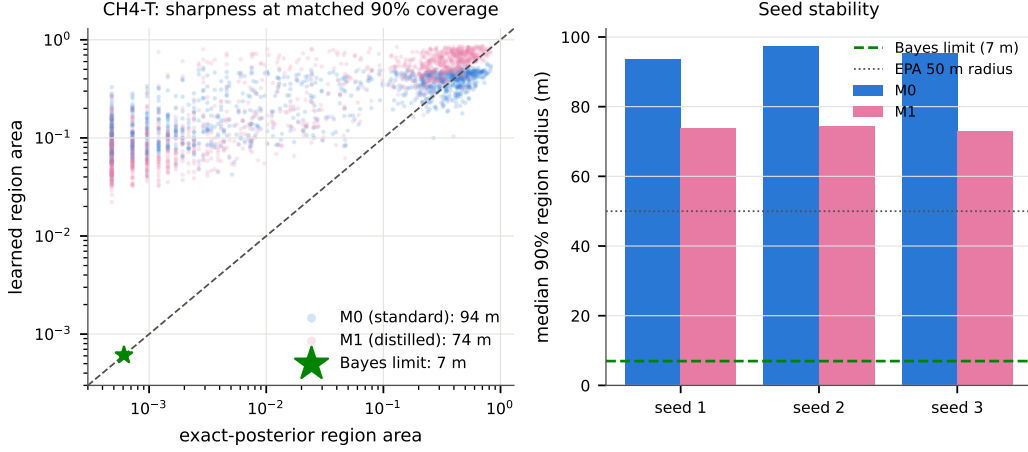


Figure 1: **CH4-T audit.** Left: learned vs. exact region area at matched 90% coverage; M0 (blue) and M1 (magenta) sit far above the diagonal, M1 nearer it; the star is the 7 m Bayes limit. Right: median region radius across three seeds — M1 strictly below M0, both calibrated, against the Bayes limit and the EPA 50 m radius.

4 Results

Calibrated is not actionable. Conformal M0 covers at 0.89, but its median 90% region has a 94 m equivalent radius — against a 7 m Bayes limit (Figure 1, left; the cloud sits far above the diagonal). The data support a near-pinpoint, but a standard amortized softmax delivers a region an order of magnitude wider in radius, because a blurry classifier cannot reproduce a near-delta posterior. This is the amortization gap, and it is large precisely because time-of-arrival makes the posterior sharp.

Distillation sharpens, robustly. Supervising the same network with the tempered exact posterior (M1) tightens the median region to a 74 m radius while preserving calibration (0.90). The improvement is stable across three seeds — M1 (73–74 m) never overlaps M0 (94–98 m; Figure 1, right) — and highly significant (paired Wilcoxon on per-scenario log ratios, $p < 10^{-60}$; M1 sharper on the majority of scenarios). Point accuracy is essentially unchanged: what distillation fixes is the shape of the uncertainty, not the location estimate.

Tempering is necessary. Distilling the *raw* exact posterior fails: its median region (134 m) is worse than M0’s, because near-delta targets provide gradients dominated by a few cells the student cannot match. Tempering the teacher to the scale of the baseline bump is what makes the physics signal learnable — a finding that replicates the same effect we observe on two other dispersion testbeds.

Operational reading. Under CH4-T’s units the median search region shrinks from a 94 m to a 74 m radius (a $\sim 40\%$ cut in search *area*) at identical confidence and zero inference cost, moving attribution toward the EPA 50 m actionability radius (Figure 2). The residual gap to the 7 m limit is the honest cost of amortization at this capacity: the training signal, not the sensor data, is now the binding constraint, and a better *target* — not a bigger network — is what closed most of the reachable distance.

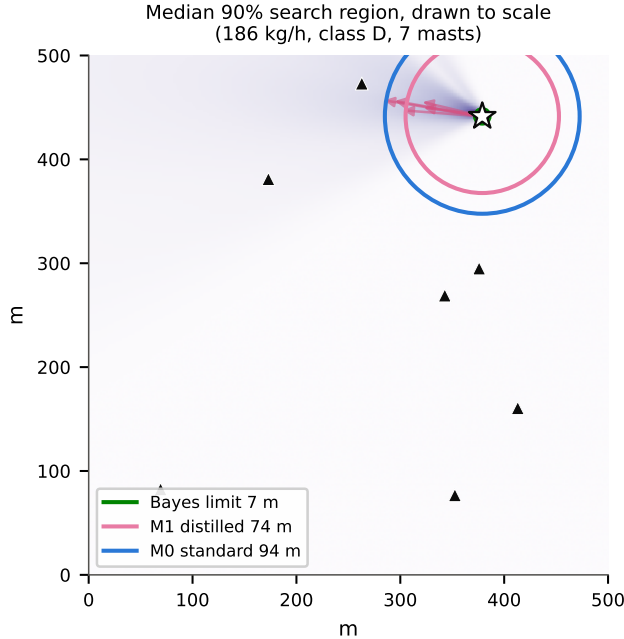


Figure 2: **One scenario, to scale.** The wind-swept plume (purple; meander arrows) and masts on the 500 m site, with the median 90% search regions drawn as equivalent-radius circles: the data permit a 7 m circle (Bayes limit, at the star), a standard localizer forces 94 m, and physics-posterior distillation trims it to 74 m.

5 Discussion

The exact posteriors earn their storage twice. As a *reference*, they show that coverage alone would certify a localizer whose regions are an order of magnitude too wide; only the exact-posterior audit, in physical units, exposes the gap and reads it against the regulatory radius. As a *training signal*, the same posteriors — tempered — sharpen the localizer substantially with no architecture, data, or inference-cost change, demonstrating that the deficiency was posterior approximation, not localization accuracy. The recipe is general to simulator-based inverse problems where a tractable posterior can be computed offline for training scenarios.

Limitations. CH4-T is an idealized testbed: a steady release, quasi-static 3-D plume sensed in 2-D, flat terrain, known stability class, fixed release height, and *measured* (known) wind. Real sites violate these; the coverage guarantee is marginal and in-distribution. The residual 74 m vs. 7 m gap means even the distilled localizer recovers a fraction of achievable sharpness, and we study the training target rather than architecture as the lever. The pathway to deployment runs through unknown/misspecified-wind robustness and recalibration on controlled-release data (e.g., a METEC-style range). We present this as evidence that the reliability machinery for trustworthy methane attribution works and is worth transferring — nothing more.

References

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